PAPER_785: Spirograph Nebula IC 418 — UQFF Planetary Nebula Fast Wind

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Framework: UQFF (Universal Quantum Field Superconductive Framework)
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Abstract

IC 418, the "Spirograph Nebula," is one of the most intricately structured planetary nebulae, located ~2,000 light-years away ($z \approx 0.0007$) in the constellation Lepus. Hubble WFPC2 imaging reveals a complex pattern of nested shells and radial spokes resembling a spirograph drawing. The central white dwarf (T_eff ~36,000 K) drives a fast stellar wind at ~1,500 km/s ($v = 1.5\times10^6$ m/s), ionizing the ejected AGB envelope. Under UQFF, the fast wind velocity ($v = 1.5\times10^6$ m/s) with standard PN B-field ($B = 10^{-5}$ T) yields g_IC418 $\approx 1.580\times10^{-2}$ m/s^2.

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1. Introduction

IC 418 represents a late-stage AGB star (~1–3 M⊙ progenitor) that ejected its outer envelope ~3,000 years ago, creating the current planetary nebula shell. The Hubble image reveals the "spirograph" interference pattern from multiple overlapping AGB pulsation shells that were ejected at slightly different angles. The central star's fast wind at ~1,500 km/s (confirmed by UV spectroscopy) is the highest drive velocity in the slow/fast wind PN interaction model. UQFF encodes this through $v = 1.5\times10^6$ m/s, which is 1.5× the LBV eruptive velocity (and thus 15× the standard HII velocity), yielding $g = 1.580\times10^{-2}$ m/s^2.

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2. Master UQFF Gravity Equation

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g_IC418(r, t) = (G × M) / r² × (1 + H(z)×t) × (1 - E_rad) × (1 + f_TRZ) + a_EM
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2.1 Parameters

Parameter	Symbol	Value	Source
Nebula mass (envelope)	M	~0.6 M⊙ = 1.193×10^{30} kg	Hubble
Nebula radius	r	1×10^{16} m (~0.1 pc)	Hubble
Age	t	~3,000 yr = 9.468×10^{10} s	Expansion age
E_rad	—	0.20	EUV photoionization loss
Redshift	z	0.0007	Distance
v_EM	v	1.5×10^6 m/s	Central star fast wind
B_EM	B	10^{-5} T	PN field
f_TRZ	—	0.05	UQFF

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3. Long-Form Derivation

Step 1: Base Gravitational Term

$g\_grav = 6.6743e-11 \times 1.193e30 / (1e16)^2 = 7.962e-13\ m/s^2$

Step 2: Cosmic Expansion Factor

$H(z) \times t \approx$ negligible ($2.268e-18 \times 9.468e10 \approx 2.15e-7$); factor ≈ 1.0000002

Step 3: Radiation Energy Loss (EUV Photoionization)

$E\_rad = 0.20;\ 1 - E\_rad = 0.80$

Step 4: Time-Reversal Correction

$f\_TRZ = 0.05;\ 1 + f\_TRZ = 1.05$

Step 5: Gravitational Total

$g\_grav\_total = 7.962e-13 \times 1.0 \times 0.80 \times 1.05 = 6.689e-13\ m/s^2$

Step 6: Aether EM Correction (Fast Wind)

$v = 1.5\times10^6$ m/s, $B = 10^{-5}$ T

$a\_EM = (1.602e-19 \times 1.5e6 \times 1e-5 / 1.673e-27) \times 11 \times 1e-12 = 1.580e-2\ m/s^2$

Step 7: Final Solution

$g\_IC418 = 6.689e-13 + 1.580e-2 \approx 1.580e-2\ m/s^2$

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4. Physical Interpretation

IC 418's result ($g = 1.580\times10^{-2}$) falls precisely between the standard HII (1.053×10^{-3}) and LBV eruptive (1.053×10^{-2}) regimes. The 1.5× velocity multiplier (1.5×10^6 vs. 1.0×10^6 m/s) directly yields a 1.5× larger result. IC 418 establishes the "fast wind PN" UQFF subcategory at $v = 1.5\times10^6$ m/s and will be shared with NGC 6307+7027 (PAPER_788) and M57 (PAPER_791) as the canonical planetary nebula fast-wind value.

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5. Conclusions

UQFF applied to IC 418 Spirograph Nebula yields $g\_IC418 \approx 1.580\times10^{-2}$ m/s^2, establishing the planetary nebula fast-wind UQFF parameter class. The 1.5×10^6 m/s central star wind velocity is directly observed in UV spectroscopy, providing an observational anchor for the PN fast-wind UQFF subcategory.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.097 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\mathrm{DVP}} = 13, \quad n_{\mathrm{channel}} = 6/26$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 13$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits its sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework

work

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t-t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.097	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 13$	✓ Sub-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.